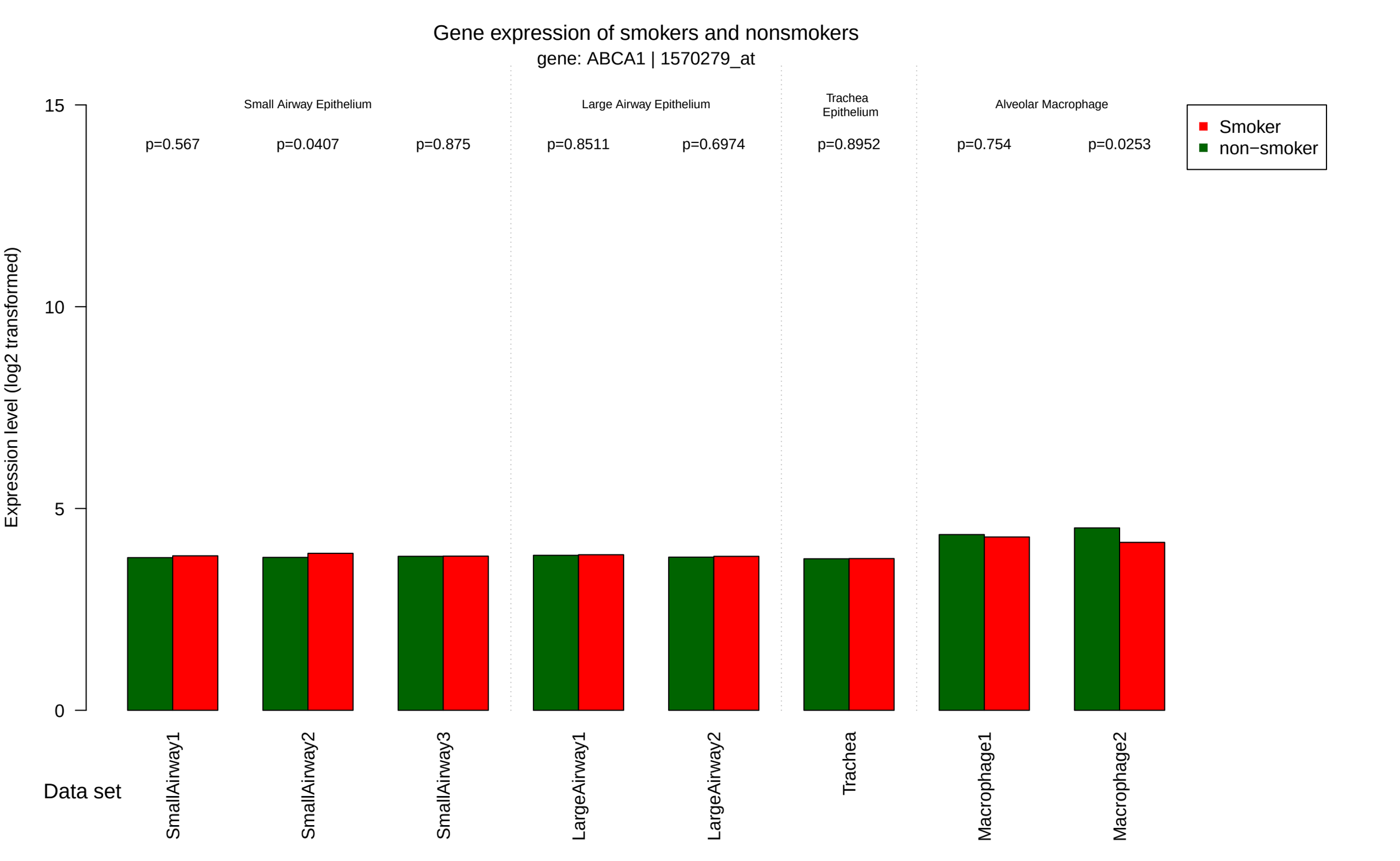
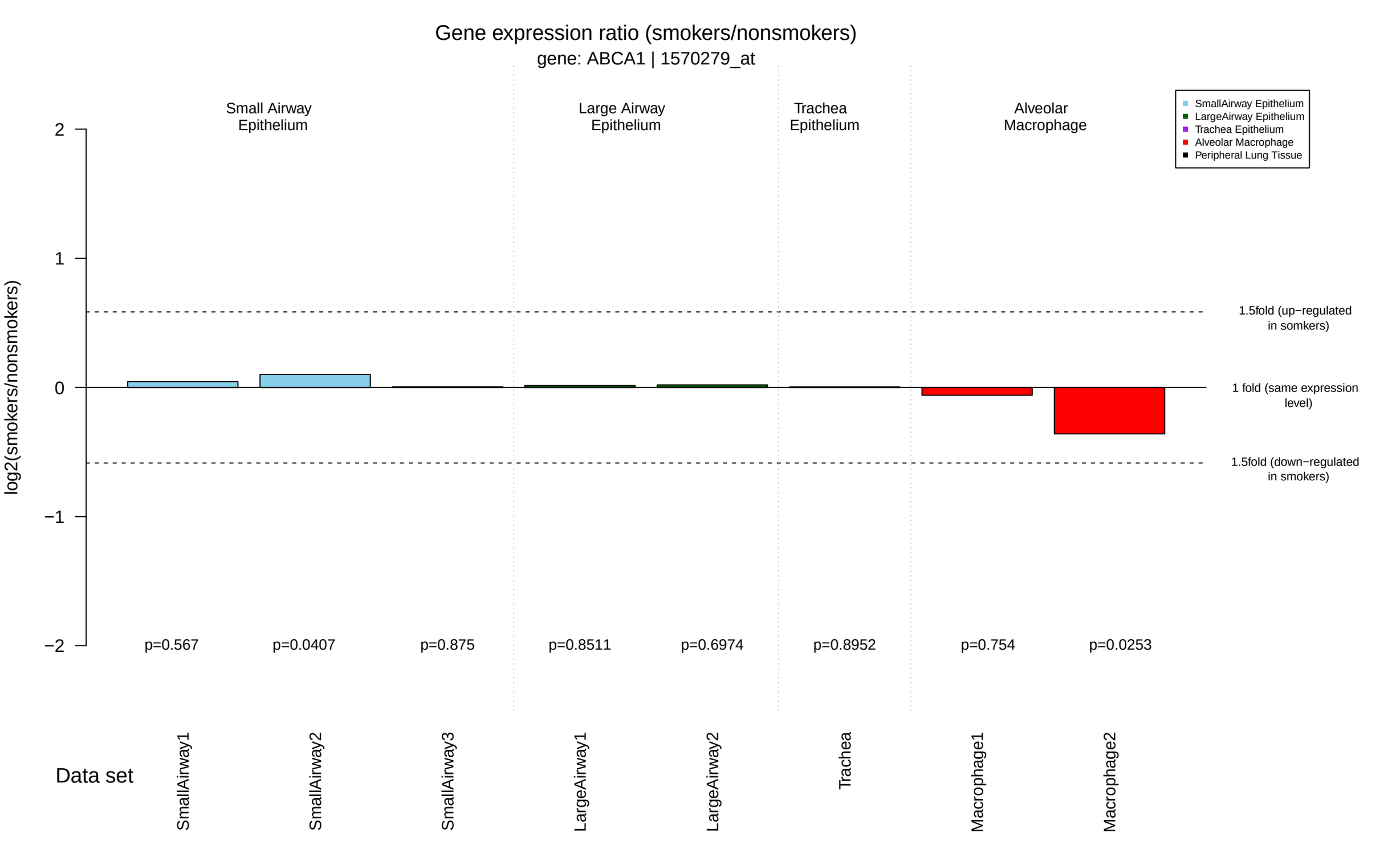
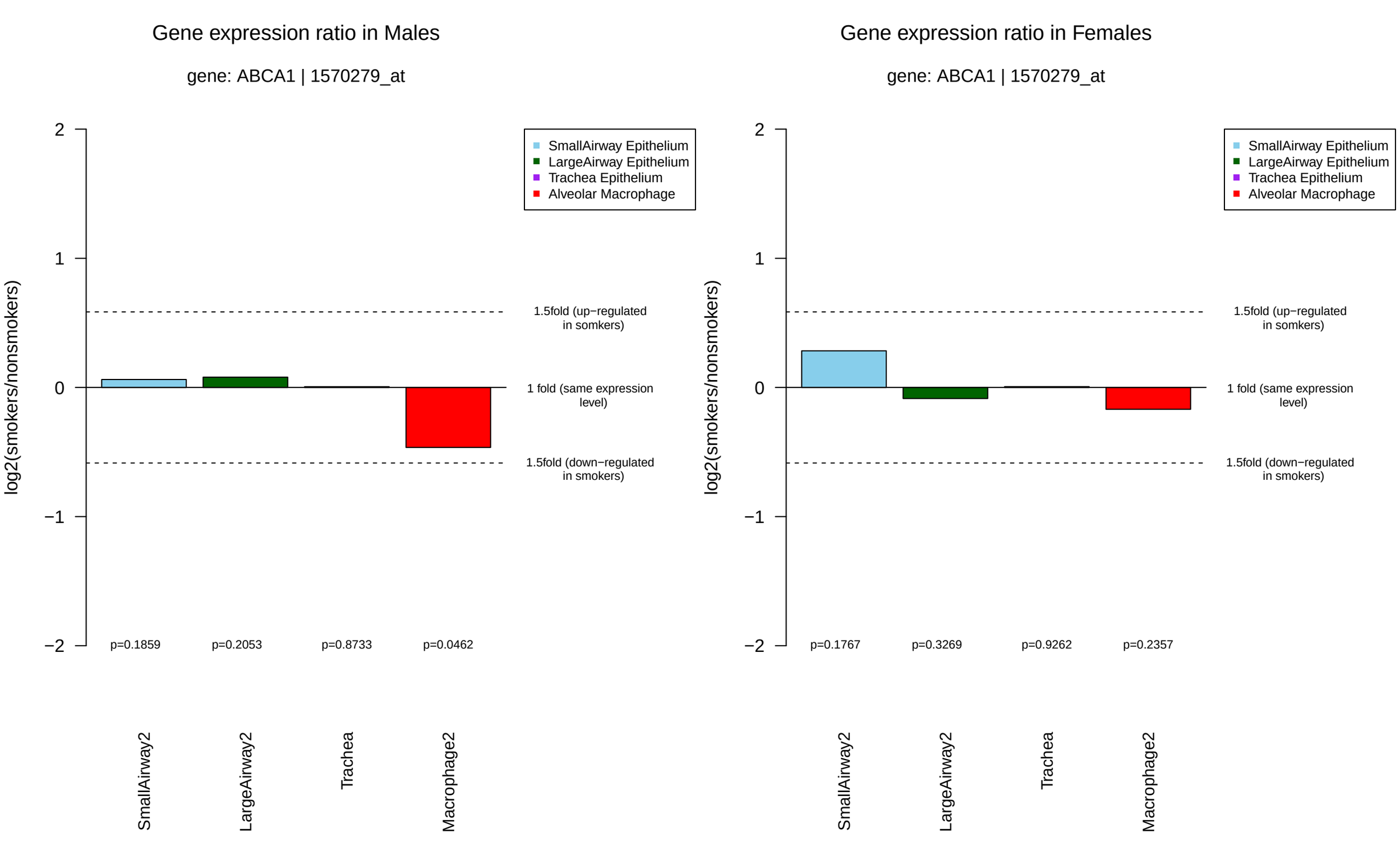
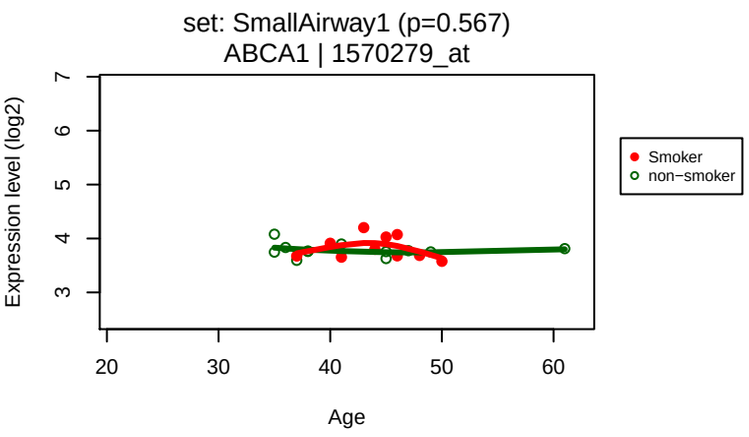
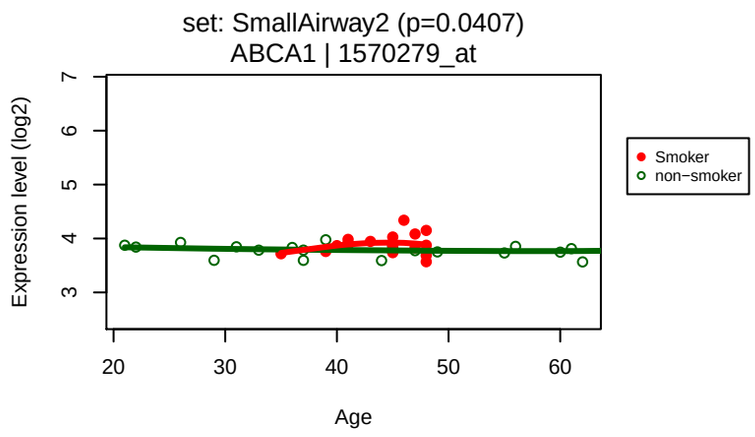
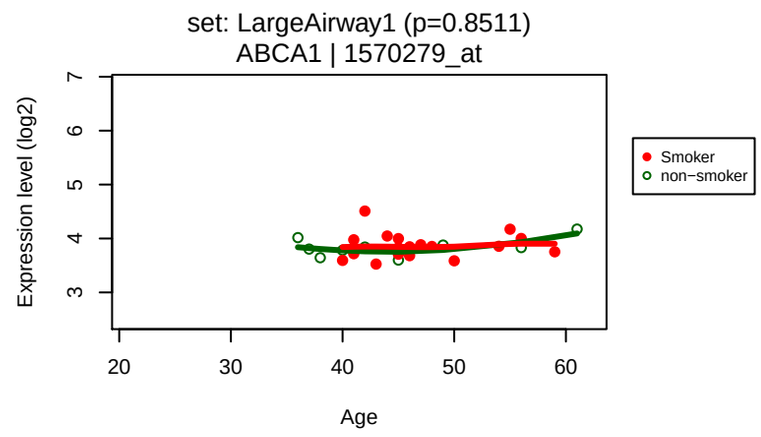

Gene expression of smokers and nonsmokers
gene: ABCA1 | 1570279_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.567
p=0.0407
p=0.875
p=0.8511
p=0.6974
p=0.8952
p=0.754
p=0.0253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

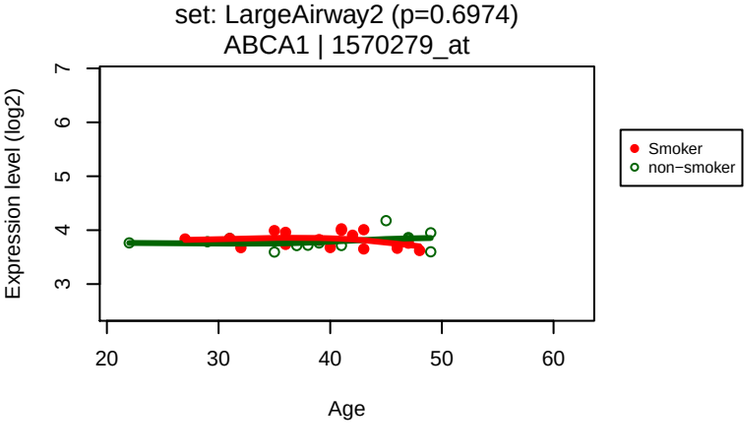

Gene expression ratio (smokers/nonsmokers)
gene: ABCA1 | 1570279_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.567
p=0.0407
p=0.875
p=0.8511
p=0.6974
p=0.8952
p=0.754
p=0.0253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

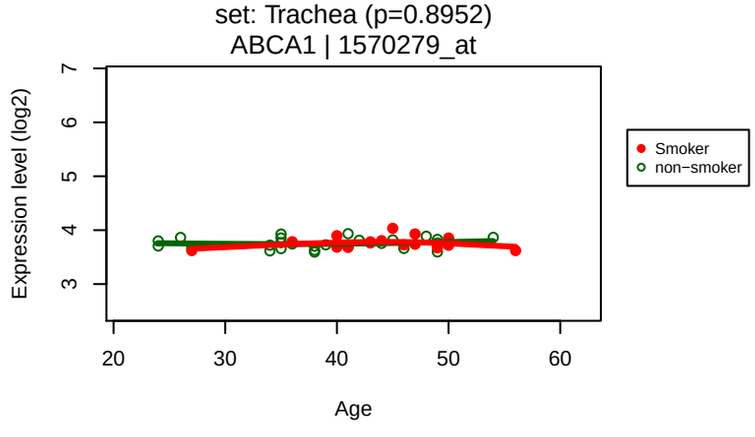

Gene expression ratio in Males
gene: ABCA1 | 1570279_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1859
p=0.2053
p=0.8733
p=0.0462
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCA1 | 1570279_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1767
p=0.3269
p=0.9262
p=0.2357
SmallAirway2
LargeAirway2
Trachea
Macrophage2

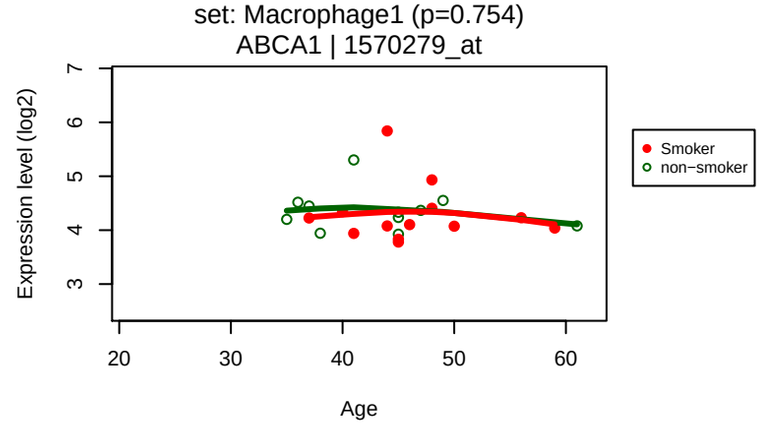

set: SmallAirway1 (p=0.567)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker

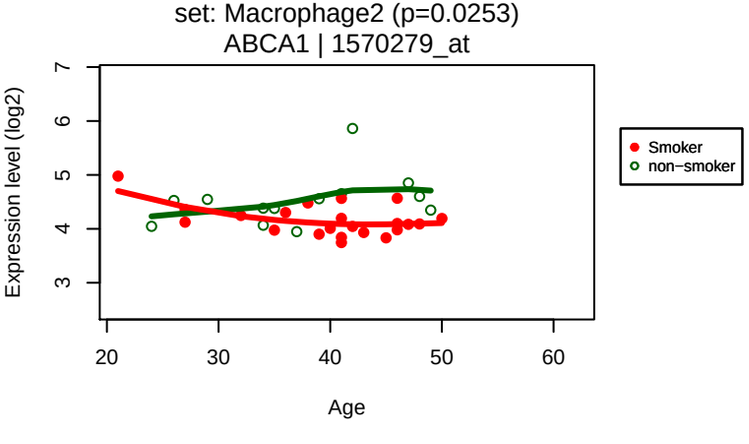

set: SmallAirway2 (p=0.0407)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker

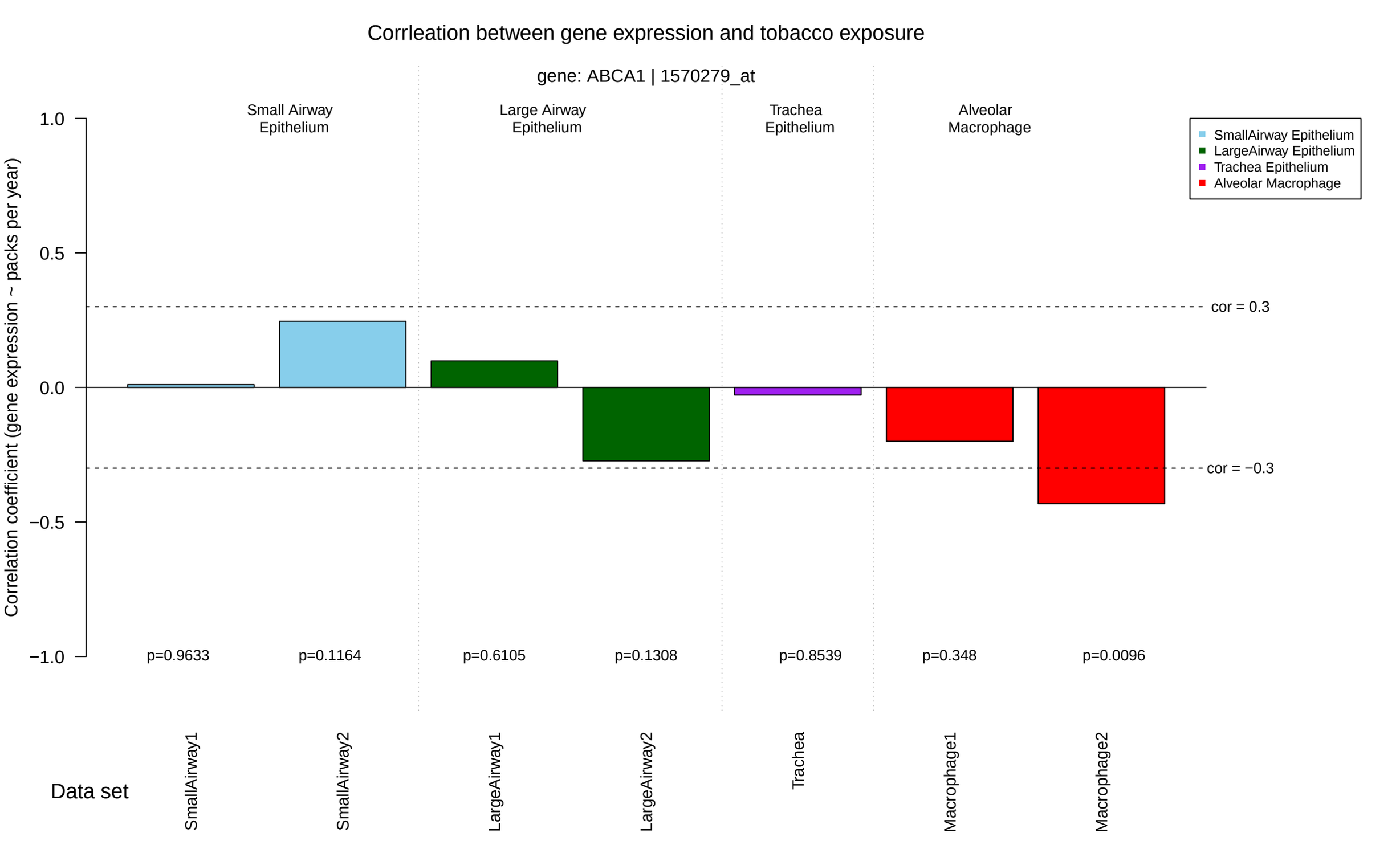

set: LargeAirway1 (p=0.8511)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6974)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8952)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.754)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0253)
ABCA1 | 1570279_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCA1 | 1570279_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9633
p=0.1164
p=0.6105
p=0.1308
p=0.8539
p=0.348
p=0.0096
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2